



## Relations & Cryptography

→ RSA Cryptography: Uses Modulo Arithmetics to de- & encrypt

$C = \text{Ciphertext}$   
 $M = \text{Plaintext}$   
 $e/s = \text{Public Key}$   
 $d = \text{Private Key}$

$M = (M^e)^d \bmod S$   
 $C = M^e \bmod S \longrightarrow M = C^d \bmod S$

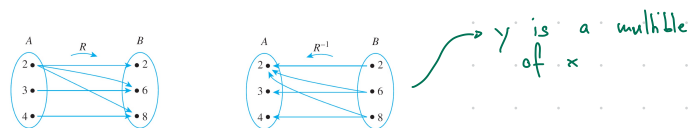
$S = p \cdot q$  |  $p, q$  are Primes  
 $e$  is relatively prime to  $(p-1)(q-1)$   
 $d$  is positive inverse to  $e \bmod (p-1)(q-1)$

only known by recipient

→ Relations: More general than Functions

$x R y \rightarrow x$  is related to  $y$   
 $x \not R y \rightarrow x$  is not related to  $y$   
 $R$  of set  $A$  to set  $B$  is a subset of cartesian Product  
 $R = \{(x, y) \in A \times B \mid x R y\}$   
 $R^{-1} = \{(y, x) \in A \times B \mid (x, y) \in R\}$

$x R y \Leftrightarrow x \mid y$   
 (It reads as "x divides y" or "x is a factor of y" or "y/x is an integer")



Relations can also be defined on one Set:  $R = \{(x, y) \in A^2 \mid x R y\}$

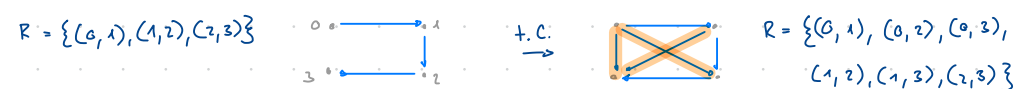
Relation on Finite Set's are often shown in directed graphs

Properties of Relations:

- Reflexivity:  $\forall x \in A : x R x$
- Symmetry:  $\forall (x, y) \in A^2 : x R y \Leftrightarrow y R x$
- Transitivity:  $\forall x, y, z \in A^2 : (x R y \ \& \ y R z) \rightarrow x R z$

Transitive Closure of Relation  $R \rightarrow R^t$

$A$  Relation may need some additional ordered Pairs to pass transitivity  
 Adding the minimum of these pairs to ensure transitivity is called transitive closure



Partitions: Mutually disjoint subsets of  $A$  whose Union is  $A$

A Relation induced by a partition: every element of partition is linked to themselves and every other element

Every Relation induced by partition is an equivalence Relation

Equivalence Class: all elements in equivalence Relation of Set  $A$  related to element  $a$

$[a] = \{x \in A \mid x R a\}$  → every  $x$  is called class representative

Congruence Modulo Relation: Congruence Modulo  $u$

$m \equiv n \pmod{d} \Leftrightarrow d \mid (m-n) \Leftrightarrow d$  divides  $(m-n)$

$m \bmod d = n \bmod d$

example:  $R = \{(x, y) \in \mathbb{Z}^2 \mid 3 \mid (x-y)\}$  → find distinct equivalence Classes:

$[0] = \{x \in \mathbb{Z} \mid x = 3k + 0 \ k \in \mathbb{Z}\}$  /  $[1] = \{x \in \mathbb{Z} \mid x = 3k + 1 \ k \in \mathbb{Z}\}$  /  $[2] = \{x \in \mathbb{Z} \mid x = 3k + 2 \ k \in \mathbb{Z}\}$

Properties:

- $(a+b) \bmod u = (a \bmod u + b \bmod u) \bmod u$
- $(a \cdot b) \bmod u = (a \bmod u \cdot b \bmod u) \bmod u$
- $(a^b) \bmod u = (a \bmod u)^b \bmod u$

Useful operations:  $(x^4) \bmod u = (x^2 \bmod u)^2 \bmod u$

## Modulo Arithmetics

Finding the Inverse of Modulo  $u$

$a \bmod u \xrightarrow{a^{-1}} (a \cdot x) \bmod u = 1$

$x$  only exists if  $\text{gcd}(a, u) = 1$

extended Euclidean Algo finds inverse

① Confirm  $\text{gcd}(a, u) = 1$

② find  $s, t$  so that  $sa + tu = 1$

$s$  is inverse (if negative → add  $u$ )

Bsp:  $23 \bmod 210 \rightarrow$  inverse  $x?$

| euclidean                | extension              |
|--------------------------|------------------------|
| $\text{gcd}(210, 23)$    |                        |
| $= \text{gcd}(23, 3)$    | $3 = 210 - 9 \cdot 23$ |
| $= \text{gcd}(3, 2)$     | $2 = 23 - 7 \cdot 3$   |
| $= \text{gcd}(2, 1)$     | $1 = 3 - 1 \cdot 2$    |
| $= \text{gcd}(1, 0) = 1$ |                        |

$$\begin{aligned}
 1 &= 3 - 1 \cdot 2 = (210 - 9 \cdot 23) - 1(23 - 7 \cdot 3) \\
 1 &= (210 - 9 \cdot 23) - (23 - 7 \cdot 23) \\
 1 &= 8 \cdot 210 - 72 \cdot 23 \\
 \rightarrow x &= -72 \rightarrow x = -72 + 210 = 138 \\
 (138 \cdot 23) \bmod 210 &= 1
 \end{aligned}$$

→ BSP RSA Encryption:

Two random Primes:  $p = 7 \quad q = 5$

$S = pq = 35$

$e$  must be an int relatively prime to  $(p-1)(q-1)$

$\text{gcd}(e, 24) = 1 \rightarrow e = 11$

→ Public Keys:  $S = 35 \quad e = 11$

$d: \quad ed \equiv 1 \pmod{(p-1)(q-1)}$

$ed \equiv 1 \pmod{24}$

→ inverse

$$\begin{aligned}
 e \bmod ((p-1)(q-1)) &= 11 \bmod 24 \\
 \text{gcd}(11, 2) &= 24 - 2 \cdot 11 \\
 \text{gcd}(2, 1) &= 11 - 5 \cdot 2 \\
 \text{gcd}(1, 0) &= 1
 \end{aligned}$$

$$\begin{aligned}
 1 &= 11 - 5 \cdot (24 - 2 \cdot 11) \\
 1 &= -5 \cdot 24 + 11 \cdot 11 \\
 \rightarrow \text{Private key } d &= 11
 \end{aligned}$$

Encryption: Plain text = 12

$$\begin{aligned}
 C &= M^e \bmod S = (12)^{11} \bmod 35 \\
 C &= ((12^4 \bmod 35) \cdot (12^4 \bmod 35) \cdot (12^3 \bmod 35)) \bmod 35 \\
 C &= ((12)(4)(16)^2 \bmod 35) \bmod 35 \\
 C &= ((12)(4)(16)^2 \bmod 35) \bmod 35 = 3
 \end{aligned}$$

Decryption:  $M = C^d \bmod S$

$$\begin{aligned}
 M &= (3^{11}) \bmod 35 = ((3^4 \cdot 35) \cdot (3^2 \cdot 35) \cdot (3^4 \cdot 35) \cdot 3) \bmod 35 \\
 &= ((3)(3)(11)^2 \cdot 35) \bmod 35 = (2 \cdot 3 \cdot 16) \bmod 35 = 12
 \end{aligned}$$

## Graphs and Trees

Graphs are a set of vertices connected by edges

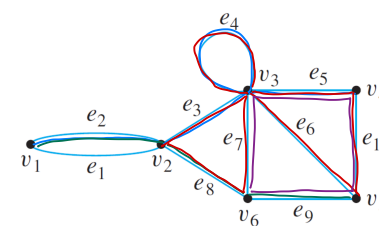
A complete Graph with  $n$  vertices has  $\frac{n(n-1)}{2}$  edges

The degree of a vertex is number of edges that end or start at  $v$

Total degree of graph = sum of degree of all  $v_i$  →  $\deg(v) = 5$

These are 4 types of Routes in a Graph:

- Trails: Routes that don't have a repeated edge
- Paths: Trails that don't have a repeated vertices
- Circuits: Trails that end at its starting vertex
- Simple Circuits: Circuits that don't repeat a vertex

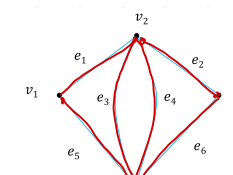


A Graph is connected if there is a path from every vertex to every vertex

connected components are subpaths with no connection

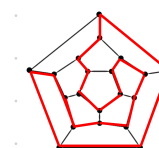
An euler circuit passes

through every vertex at least once and through every edge exactly once  
 can only appear if every vertex has even degree and the graph is connected



A Hamiltonian Circuit visits

every vertex exactly once and no edge more than once  
 simple circuit that passes through all vertices



Matrix representation of Graph: read from L to R.

$$A = \begin{bmatrix} v_1 & v_2 \\ v_2 & v_1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Matrices of undirected Graphs are symmetric

Computing the number of distinct routes with length  $n$   
 $A^n = \#$  of routes connecting  $v_i$  with  $v_j$

Isomorphism of Graphs

Two Graphs are isomorphic if they can be mapped on to each other

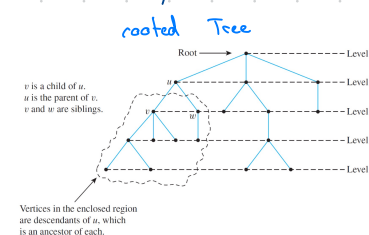
There is no efficient Algorithm known

Trees are Graphs without any Circuits

max Level = height

Binary Trees: each vertex has max 2 Children

Oct Trees: exactly 8 children per vertex



A Spanning Tree connects every vertex of a Graph

Minimum Spanning Tree → ST with minimum weights

Two Algo's to find MST:

① Kruskal's Algo: - Mark edge with min weight  
 - Add next min edge if no circuit  
 - Repeat until its impossible to add  
 - Faster for sparse Graphs

② Prim's Algo: - Take arbitrary vertex  
 - Take min edge, make visited edge  
 - Take min edge of all visited vertices  
 - Repeat until all  $v$  are visited  
 - Faster for dense Graphs

Dijkstra's Algo: for finding the shortest path

- Define starting vertex  $S$   
 - Assign to all adjacent vertices cost  
 - Check all visited vertices and search the unvisited with lowest cost from  $S$   
 - repeat until all  $v$  are visited



Topological Map for Robots:

Each node represents a Location  
 Each edge a physical connection

Volumetric Map for Robots:

A Grid allows robot to know where objects are

SLAM: Simultaneous Localization and Mapping